LENOVO | DATA CENTER STORAGE

Trident 25.06 documentation

For Lenovo DM/DG/DS-Series storage running ONTAP

Trident 25.06.2 · Container Storage Interface (CSI) provisioner

Document revision: 25.06-v1 (preview)

About this deliverable

Scope of this preview build. This PDF mirrors the structure, formatting, and chapter order of the upstream Trident 25.06 full-site documentation, with chapters and content that do not fit the Lenovo DM/DG/DS-Series portfolio removed or rewritten per the Lenovo redwash playbook. Version strings were taken from the canonical `trident-2506` documentation set and are already correct for the 25.06.2 patch Lenovo ships.

This build contains fully redwashed content for the front-matter and reference chapters that fit on-premises Lenovo storage (Release notes → Get started → Requirements → Use Trident / worker-node prep → ONTAP SAN driver overview), plus the fully-specified Knowledge & support and Legal notices chapters. Continuation points are marked where the remaining retained chapters (detailed ONTAP NAS/SAN backend options, install procedures, manage and monitor, data protection, security, and the object/CRD reference) extend in subsequent passes.

Contents (Lenovo deliverable)

Release notes

What’s new

Release notes provide information about new features, enhancements, and bug fixes in this version of Trident.

The `tridentctl` binary for Linux provided in the installer ZIP file is the tested and supported version. The macOS binary provided in the `/extras` part of the ZIP file is not tested or supported.

What’s new in 25.06.2

Fixes

- Kubernetes: Fixed a critical issue where incorrect iSCSI devices were discovered when detaching volumes from Kubernetes nodes.

Changes in 25.06.1

Fixes

- Kubernetes:
  - Fixed an issue where NQNs were not checked before being unmapped from subsystems.
  - Fixed an issue where multiple attempts to close a LUKS device led to failures in detaching volumes.
  - Fixed iSCSI volume unstage when the device path has changed since its creation.
  - Block cloning of volumes across storage classes.
- OpenShift: Fixed an issue where iSCSI node prep failed with OCP 4.19.

Changes in 25.06

Enhancements — Kubernetes

- Added support for CSI Volume Group Snapshots with v1beta1 Volume Group Snapshot Kubernetes APIs for the ONTAP-SAN iSCSI driver. See *Work with volume group snapshots*.
- Added support for Lenovo DS-Series storage systems for NVMe/TCP in addition to iSCSI. See *ONTAP SAN configuration options and examples*.
- Added secure SMB support for ONTAP-NAS and ONTAP-NAS-Economy volumes. Active Directory users and groups may now be used with SMB volumes for enhanced security. See *Enable secure SMB*.
- Enhanced Trident node concurrency for higher scalability on node operations for iSCSI volumes.
- Added `--allow-discards` when opening LUKS volumes to allow discard/TRIM commands for space reclamation.
- Enhanced performance when formatting LUKS-encrypted volumes; enhanced LUKS cleanup for failed but partially formatted LUKS devices.
- Enhanced Trident node idempotency for NVMe volume attach and detach.
- Added `internalID` field to the Trident volume config for the ONTAP-SAN-Economy driver.
- Added support for volume replication with SnapMirror for NVMe backends. See *Replicate volumes using SnapMirror*.

Experimental enhancements (not for use in production)

- [Tech Preview] Enabled concurrent Trident controller operations via the `--enable-concurrency` feature flag, allowing controller operations to run in parallel. Currently supports limited parallel workflows with the ONTAP-SAN driver (iSCSI and FCP protocols).

Fixes

- Fixed an issue with CSI `NodeExpandVolume` where multipath devices could be left with incongruent sizes when underlying SCSI disks are unavailable.
- Fixed failure to clean up duplicate export policies for ONTAP-NAS and ONTAP-NAS-Economy drivers.
- Fixed deployment issue when installing Trident using Kustomize (Issue #831).
- Fixed missing export policies for PVCs created from snapshots (Issue #1016).
- Fixed issue with ONTAP-NAS-Economy volumes expanding up to 300 TB despite resize failures.
- Fixed issue where clone split operations were being done synchronously when using the ONTAP REST API.

Deprecations

- Kubernetes: Updated minimum supported Kubernetes to v1.27.

> **Redwash notes for this chapter:** Trident Protect "What's new" subsections were removed (Trident Protect deferred for 25.06). Bullets specific to Azure NetApp Files (manual QoS tech preview), Google Cloud NetApp Volumes (NFSv3 default fix), and SolidFire (clone-timeout fixes, the 25.06.1 SolidFire upgrade caution) were eliminated as off-product. All `25.02` -and-earlier "Changes in…" sections were stripped (25.06-only deliverable).

Known issues

Known issues identify problems that might prevent you from using this release of the product successfully.

- When upgrading a Kubernetes cluster from 1.24 to 1.25 or later that has Trident installed, you must update `values.yaml` to set `excludePodSecurityPolicy` to `true` (or add `--set excludePodSecurityPolicy=true` to the `helm upgrade` command) before you can upgrade the cluster.
- Trident enforces a blank `fsType` (`fsType=""`) for volumes that do not specify `fsType` in their StorageClass. For iSCSI volumes, you are required to set `fsType` on your StorageClass when enforcing an `fsGroup` using a Security Context.
- When using a backend across multiple Trident instances, each backend configuration file should use a different `storagePrefix` value for ONTAP backends. Trident cannot detect volumes other instances created.
- There should be only one instance of Trident per Kubernetes cluster.
- To perform online space reclamation for iSCSI PVs, the underlying OS on the worker node might require the `discard` mount option (for example, RHEL/RHCOS instances); include the `discard` mountOption in your StorageClass.
- ONTAP cannot concurrently provision more than one FlexGroup at a time unless the set of aggregates is unique to each provisioning request.
- When using Trident over IPv6, specify `managementLIF` and `dataLIF` in the backend definition within square brackets, for example `[fd20:8b1e:b258:2000:f816:3eff:feec:0]` . You cannot specify `dataLIF` on an ONTAP SAN backend; Trident discovers all available iSCSI LIFs and uses them to establish the multipath session.

Redwash note: The `solidfire-san` /Element OS 12.7 CHAP-algorithm known issue and the Trident Protect “Restoring Restic backups of large files” known issue were removed (off-product / deferred). The *Find more information* “Trident GitHub” link now points to `github.com/Lenovo/trident` ; the blog link was removed.

Get started

Learn about Trident

Trident is a fully-supported open source project maintained by NetApp. It is designed to help you meet your containerized application's persistence demands using industry-standard interfaces such as the Container Storage Interface (CSI).

What is Trident?

Trident enables consumption and management of storage resources on Lenovo DM/DG/DS-Series storage running ONTAP, deployed on premises. Trident is a Container Storage Interface (CSI) compliant dynamic storage orchestrator that natively integrates with Kubernetes. Trident runs as a single Controller Pod plus a Node Pod on each worker node in the cluster. Refer to *Trident architecture* for details.

> **NOTE**
> If this is your first time using Kubernetes, familiarize yourself with the Kubernetes concepts and tools.

Kubernetes integration with Lenovo storage

Lenovo DM/DG/DS-Series storage integrates with many aspects of a Kubernetes cluster, providing advanced data management capabilities that enhance the functionality, capability, performance, and availability of the Kubernetes deployment.

ONTAP

ONTAP is the multiprotocol, unified storage operating system that provides advanced data management capabilities for any application. Lenovo DM/DG/DS-Series arrays run ONTAP and are available in all-flash, hybrid, and SAN configurations. Trident supports these on-premises deployment models.

Redwash note: The upstream "What is Trident?" paragraph listing FAS/AFF/ASA, ONTAP Select, Cloud Volumes ONTAP, Element software (NetApp HCI, SolidFire), Azure NetApp Files, Amazon FSx, and Cloud Volumes Service was reduced to on-premises Lenovo storage running ONTAP. The NetApp Docker Volume Plugin (nDVP) paragraph was removed (out of Lenovo support scope). The per-product subsections for FSx, Azure NetApp Files, Cloud Volumes ONTAP, Google Cloud NetApp Volumes, Element, and NetApp HCI were removed. "maintained by NetApp" is retained as required origin attribution.

Trident architecture

Trident runs as a single Controller Pod plus a Node Pod on each worker node in the cluster. The Node Pod must run on any host where you want to potentially mount a Trident volume.

Controller pods and node pods

Trident deploys as a single Trident Controller Pod and one or more Trident Node Pods on the Kubernetes cluster and uses standard Kubernetes CSI sidecar containers to simplify the deployment of CSI plugins. Node selectors, tolerations, and taints can be used to constrain a pod to a specific or preferred node.

- The controller plugin handles volume provisioning and management, such as snapshots and resizing.
- The node plugin handles attaching the storage to the node.

Trident Controller Pod

The Trident Controller Pod is a single Pod running the CSI Controller plugin. It is:

- Responsible for provisioning and managing volumes in Lenovo storage
- Managed by a Kubernetes Deployment
- Able to run on the control-plane or worker nodes, depending on installation parameters

Trident Node Pods

Trident Node Pods are privileged Pods running the CSI Node plugin. They are:

- Responsible for mounting and unmounting storage for Pods running on the host
- Managed by a Kubernetes DaemonSet
- Required to run on any node that will mount Lenovo storage

Supported Kubernetes cluster architectures

Kubernetes cluster architecture	Supported	Default install
Single master, compute	Yes	Yes
Multiple master, compute	Yes	Yes
Master, etcd, compute	Yes	Yes
Master, infrastructure, compute	Yes	Yes

Concepts

Provisioning

Provisioning in Trident has two primary phases. The first phase associates a storage class with the set of suitable backend storage pools and occurs as preparation before provisioning. The second phase includes volume creation itself and requires choosing a storage pool from those associated with the pending volume’s storage class.

Associating backend storage pools with a storage class relies on the storage class’s requested attributes and its `storagePools`, `additionalStoragePools`, and `excludeStoragePools` lists. When you create a storage class, Trident compares the attributes and pools offered by each backend to those requested by the storage class, adding suitable pools to the set for that class.

When you create a volume, Trident first gets the set of storage pools for that volume’s storage class and, if you specify a protocol, removes pools that cannot provide the requested protocol (for example, an ONTAP NAS backend cannot provide a block-based volume). Trident randomizes the order of the resulting set and iterates through it, attempting to provision the volume on each pool in turn.

Volume snapshots

- For the `ontap-nas` and `ontap-san` drivers, each Persistent Volume (PV) maps to a FlexVol volume. As a result, volume snapshots are created as ONTAP snapshots. ONTAP snapshot technology delivers stability, scalability, recoverability, and performance, and is efficient in both creation time and storage space.
- For the `ontap-nas-flexgroup` driver, each PV maps to a FlexGroup. Volume snapshots are created as ONTAP FlexGroup snapshots.
- For the `ontap-san-economy` driver, PVs map to LUNs created on shared FlexVol volumes. VolumeSnapshots are achieved by performing FlexClones of the associated LUN. ONTAP FlexClone technology makes it possible to create copies of even the largest datasets almost instantaneously.
- When working with the `ontap-nas` and `ontap-san` drivers, ONTAP snapshots are point-in-time copies of the FlexVol and consume space on the FlexVol itself. When a VolumeSnapshot created through Kubernetes is deleted, Trident deletes the associated ONTAP snapshot.

With Trident, you can use VolumeSnapshots to create new PVs from them, using FlexClone technology for supported ONTAP backends. When creating a PV from a snapshot, the backing volume is a FlexClone of the snapshot’s parent volume.

Redwash note: The `solidfire-san` snapshot bullet (Element software/NetApp HCI cluster, Element snapshots, Element volume clones) was removed. References to `gcp-cvs` and `azure-netapp-files` drivers and to “CVS backends” were dropped; “NetApp snapshot” → “ONTAP snapshot”.

Virtual storage pools

Virtual pools provide a layer of abstraction between Trident storage backends and Kubernetes StorageClasses. They allow an administrator to define aspects such as location, performance, and protection for each backend in a common, backend-agnostic way without making a StorageClass specify which physical backend, pool, or type to use.

The storage administrator can define virtual pools on any Trident backend in a JSON or YAML definition file. Any aspect specified outside the virtual pools list is global to the backend and applies to all virtual pools, while each virtual pool may override backend-global aspects. The administrator defines one or more labels (`key:value` pairs) per virtual pool; a StorageClass identifies which virtual pool to use by referencing labels within a `selector` parameter.

Operator	Example	A pool’s label value must:
`=`	`performance=premium`	Match
`!=`	`performance!=extreme`	Not match
`in`	`location in (east, west)`	Be in the set of values
`notin`	`performance notin (silver, bronze)`	Not be in the set of values
`<key>`	`protection`	Exist with any value
`!<key>`	`!protection`	Not exist

Volume access groups — removed

Removed concept. The upstream “Volume access groups” concept page is a SolidFire/Element feature and has been removed from the Lenovo deliverable and from the Get started sidebar (`vol-access-`

`groups.adoc → TO_DELETE/`). Lenovo DM/DG/DS-Series SAN access uses CHAP and per-node igroups, covered under the ONTAP SAN driver.

Quick start for Trident

You can install Trident and start managing storage resources in a few steps. Before getting started, review *Trident requirements*.

1. **Prepare the worker node.** All worker nodes in the Kubernetes cluster must be able to mount the volumes you provision for your pods.
2. **Install Trident.** Trident offers several installation methods and modes optimized for a variety of environments.
3. **Create a backend.** A backend defines the relationship between Trident and a storage system — how Trident communicates with it and how it should provision volumes.
4. **Create a Kubernetes StorageClass.** The StorageClass object specifies Trident as the provisioner and lets you provision volumes with customizable attributes.
5. **Provision a volume.** Create a PersistentVolume (PV) and a PersistentVolumeClaim (PVC) that uses the configured StorageClass, then mount the PV to a pod.

Redwash note: The “For Docker, refer to Trident for Docker” pointer was removed (Docker out of scope).

Requirements

Critical information about Trident

- Kubernetes 1.34 is now supported in Trident. Upgrade Trident prior to upgrading Kubernetes.
- Trident strictly enforces the use of multipathing configuration in SAN environments, with a recommended value of `find_multipaths: no` in the `multipath.conf` file. Use of a non-multipathing configuration or of `find_multipaths: yes` or `find_multipaths: smart` results in mount failures.

Supported frontends (orchestrators)

- Anthos On-Prem (VMware) and Anthos on bare metal 1.16
- Kubernetes 1.27 – 1.34
- OpenShift 4.12, 4.14 – 4.19 (to use iSCSI node preparation with OpenShift 4.19, the minimum supported Trident version is 25.06.1)
- Rancher Kubernetes Engine 2 (RKE2) v1.27.x – 1.34.x (qualified on RKE2 v1.28.5+rke2r1)

Trident also works with other self-managed Kubernetes offerings, including Mirantis Kubernetes Engine (MKE) and the VMware Tanzu portfolio. Trident and ONTAP can be used as a storage provider for KubeVirt.

Supported backends (storage)

To use Trident, you need one or more of the following supported backends:

- Lenovo DM/DG/DS-Series storage running ONTAP, on premises (iSCSI, NVMe/TCP, and FC), under ONTAP versions in full or limited support.

Redwash note: Removed backend rows: Amazon FSx for NetApp ONTAP, Azure NetApp Files, Cloud Volumes ONTAP, Google Cloud NetApp Volumes, NetApp All SAN Array (ASA), and NetApp HCI/Element. The cloud-K8s product list (GKE, EKS, AKS) was dropped from the frontends paragraph; MKE and VMware Tanzu retained as self-managed offerings.

Trident support for KubeVirt and OpenShift Virtualization

Trident supports the following ONTAP drivers for KubeVirt and OpenShift Virtualization:

- `ontap-nas`
- `ontap-nas-economy`
- `ontap-san` (iSCSI, FCP, NVMe over TCP)
- `ontap-san-economy` (iSCSI only)

For live-migration workflows where RWX access mode is required, these combinations are supported: NFS + `volumeMode=Filesystem`; iSCSI + `volumeMode=Block`; NVMe/TCP + `volumeMode=Block`; FC + `volumeMode=Block`.

Feature requirements

Feature	Kubernetes version	Feature gates required?
Trident	1.27 – 1.34	No
Volume Snapshots	1.27 – 1.34	No
PVC from Volume Snapshots	1.27 – 1.34	No
iSCSI PV resize	1.27 – 1.34	No
ONTAP Bidirectional CHAP	1.27 – 1.34	No
Dynamic Export Policies	1.27 – 1.34	No
Trident Operator	1.27 – 1.34	No
CSI Topology	1.27 – 1.34	No

Tested host operating systems

- Red Hat Enterprise Linux CoreOS (RHCOS) versions supported by OpenShift Container Platform on AMD64 and ARM64
- Red Hat Enterprise Linux (RHEL) 8 or later on AMD64 and ARM64 (NVMe/TCP requires RHEL 9 or later)
- Ubuntu 22.04 LTS or later on AMD64 and ARM64
- Windows Server 2022
- SUSE Linux Enterprise Server (SLES) 15 or later

Container images and corresponding Kubernetes versions

For air-gapped installations, the following container images are needed to install Trident 25.06.2. Use `tridentctl images` to verify the list.

Kubernetes versions	Container image

v1.27.0 – v1.34.0	`docker.io/netapp/trident:25.06.2`
	`docker.io/netapp/trident-autosupport:25.06`
	`registry.k8s.io/sig-storage/csi-provisioner:v5.2.0`

	`registry.k8s.io/sig-storage/csi-attacher:v4.8.1`
	`registry.k8s.io/sig-storage/csi-resizer:v1.13.2`
	`registry.k8s.io/sig-storage/csi-snapshotter:v8.2.1`
	`registry.k8s.io/sig-storage/csi-node-driver-registrar:v2.13.0`
	`docker.io/netapp/trident-operator:25.06.2` (optional)

NOTE

The `docker.io/netapp/trident*` and `registry.k8s.io/sig-storage/*` image paths are kept verbatim — the Trident container images are genuinely distributed from NetApp's Docker Hub and the CSI sidecars from the Kubernetes registry. They are technically correct on Lenovo hardware.

Install Trident

Trident offers several installation methods and modes optimized for a variety of environments. Review the installation overview to ensure you have met the installation prerequisites and selected the correct option for your environment:

- **Install using the Trident operator** (manual standard mode, or with Helm)
- **Install using `tridentctl`**
- **Install using the OpenShift certified operator**

> **IMPORTANT**
> Critical information about Trident: Kubernetes 1.34 is now supported — upgrade Trident prior to upgrading Kubernetes. Trident strictly enforces multipathing configuration in SAN environments, with a recommended value of `find_multipaths: no` in `multipath.conf`; use of a non-multipathing configuration or of `find_multipaths: yes` / `smart` results in mount failures.

Install using the Trident operator (standard mode)

You can manually deploy the Trident operator to install Trident. This process applies to installations where the container images required by Trident are not stored in a private registry. If you do have a private image registry, use the process for offline deployment.

Before you begin, log in to the Linux host and verify it is managing a working, supported Kubernetes cluster and that you have the necessary privileges.

> **NOTE**
> With OpenShift, use `oc` instead of `kubectl` in all of the examples that follow, and log in as **system:admin** first by running `oc login -u system:admin`.

```
# Verify your Kubernetes version
kubectl version

# Verify cluster administrator privileges
kubectl auth can-i '*' '*' --all-namespaces

# Verify you can launch a pod that uses an image from Docker Hub and reach your storage system
kubectl run -i --tty ping --image=busybox --restart=Never --rm -- ping <management IP>
```

Step 1: Download the Trident installer package

The Trident installer package contains everything you need to deploy the Trident operator and install Trident. Download and extract the latest version of the Trident installer from the *Assets* section on Lenovo’s GitHub distribution (`github.com/Lenovo/trident`).

```
wget https://github.com/Lenovo/trident/releases/download/v25.06.2/trident-installer-25.06.2.tar.gz
tar -xf trident-installer-25.06.2.tar.gz
cd trident-installer
```

Step 2: Create the `TridentOrchestrator` CRD

Create the `TridentOrchestrator` Custom Resource Definition (CRD). Use the appropriate CRD YAML version in `deploy/crds`.

```
kubectl create -f deploy/crds/trident.netapp.io_tridentorchestrators_crd_post1.16.yaml
```

> **NOTE**
> The API group `trident.netapp.io` is a literal Kubernetes API identifier built into the product, not branding; it is kept verbatim (changing it would break the installation).

Step 3: Deploy the Trident operator

The installer provides a bundle file to install the operator and create associated objects with a default configuration. For clusters running Kubernetes 1.25 or later, use `bundle_post_1_25.yaml`; for Kubernetes 1.24, use `bundle_pre_1_25.yaml`.

```
# Create the trident namespace if it does not exist
kubectl apply -f deploy/namespace.yaml

# Create the resources and deploy the operator
kubectl create -f deploy/<bundle.yaml>

# Verify the operator, deployment, and replicasets were created
kubectl get all -n <operator-namespace>
```

> **IMPORTANT**
> There should only be **one instance** of the operator in a Kubernetes cluster. Do not create multiple deployments of the Trident operator.

Step 4: Create the `TridentOrchestrator` and install Trident

```
kubectl create -f deploy/crds/tridentorchestrator_cr.yaml

kubectl describe torc trident
...
Status:
  Current Installation Params:
    Autosupport Image:     netapp/trident-autosupport:25.06
    Trident Image:         netapp/trident:25.06.2
    Silence Autosupport:   false
  Message:                 Trident installed
  Status:                  Installed
  Version:                 v25.06.2
```

> **NOTE**
> The image paths `netapp/trident:25.06.2` and `netapp/trident-autosupport:25.06` are retained at the NetApp Docker Hub registry — the Trident and AutoSupport images are genuinely distributed from there and are technically correct on Lenovo hardware (Category A).

Verify the installation

The status of `TridentOrchestrator` changes from `Installing` to `Installed`. You can also confirm installation by reviewing pod status, or by checking the version with `tridentctl`:

```
kubectl get pods -n trident

NAME                                READY   STATUS    RESTARTS   AGE
trident-controller-7d466bf5c7-v4cpw 6/6     Running   0          1m
trident-node-linux-mr6zc            2/2     Running   0          1m
```

```
trident-operator-766f7b8658-ldzsv     1/1     Running   0          3m

./tridentctl -n trident version
+----------------+----------------+
| SERVER VERSION | CLIENT VERSION |
+----------------+----------------+
| 25.06.2        | 25.06.2        |
+----------------+----------------+
```

Install using the Trident operator with Helm

Using the Trident Helm chart you can deploy the Trident operator and install Trident in one step. The chart version for this release is `100.2506.0`.

```
helm install trident trident-operator-100.2506.0.tgz \
  --create-namespace --namespace trident --set nodePrep={iscsi}

# Review installation details
helm list -n trident
```

To change a default value during install, pass `--set`; for example, to enable debug:

```
helm install trident trident-operator-100.2506.0.tgz \
  --create-namespace --namespace trident --set tridentDebug=true
```

Install using `tridentctl`

Download and extract the installer, then install Trident in the desired namespace by running `tridentctl install`.

```
wget https://github.com/Lenovo/trident/releases/download/v25.06.2/trident-
installer-25.06.2.tar.gz
tar -xf trident-installer-25.06.2.tar.gz
cd trident-installer

./tridentctl install -n trident
```

The installation log ends with `Trident REST interface is up. version=25.06.2` and `Trident installation succeeded.` Verify with:

```
kubectl get pods -n trident

./tridentctl -n trident version
+----------------+----------------+
| SERVER VERSION | CLIENT VERSION |
+----------------+----------------+
| 25.06.2        | 25.06.2        |
+----------------+----------------+
```

Install using the OpenShift certified operator

You can install Trident on OpenShift using the certified operator from OperatorHub. RHCOS does not have iSCSI enabled and configured by default; add the `nodePrep` parameter to configure and enable both iSCSI and Multipath services on all OpenShift worker nodes.

NOTE
Beginning with OpenShift 4.19, the minimum Trident version supported for the iSCSI node-prep feature is 25.06.1.

```
apiVersion: trident.netapp.io/v1
kind: TridentOrchestrator
metadata:
  name: trident
  namespace: openshift-operators
spec:
  IPv6: false
  debug: false
  nodePrep:
  - iscsi
  imageRegistry: ''
  k8sTimeout: 30
  namespace: trident
  silenceAutosupport: false
```

Select the Trident operator from the list of installed operators and click **Create**; the Trident Orchestrator is fully installed.

Redwash notes for this chapter: GitHub release URLs were rewritten from `github.com/NetApp/trident/…/v25.06.0/…25.06.0.tar.gz` to `github.com/Lenovo/trident/…/v25.06.2/…25.06.2.tar.gz` (Lenovo mirror, 25.06.2 patch). Version strings `25.06.0` → `25.06.2`, `v25.06.0` → `v25.06.2`, `version=25.06.0` → `version=25.06.2`. The Helm AWS/EKS `cloudProvider` example and the AWS EKS Marketplace add-on path were removed (cloud, off-product). The `netapp/` container-image registry paths, the `trident.netapp.io` API group, and the `trident-operator.netapp.io` event source are kept verbatim as literal product identifiers / Category A exceptions. The “since the 21.07 release” historical qualifier was dropped.

Use Trident

Prepare the worker node

All worker nodes in the Kubernetes cluster must be able to mount the volumes you provision for your pods. To prepare the worker nodes, install NFS, iSCSI, NVMe/TCP, or FC tools based on your driver selection.

Selecting the right tools

- **NFS tools** — install if you are using `ontap-nas`, `ontap-nas-economy`, or `ontap-nas-flexgroup`.
- **iSCSI tools** — install if you are using `ontap-san` or `ontap-san-economy`.
- **NVMe tools** — install if you are using `ontap-san` for NVMe over TCP (NVMe/TCP). Lenovo recommends ONTAP 9.12 or later for NVMe/TCP.
- **SCSI over FC tools** — install if you are using `ontap-san` with `sanType fcp`. SCSI over FC is supported on OpenShift and KubeVirt environments. iSCSI self-healing does not apply to SCSI over FC.

NFS volumes

Install the NFS tools using the commands for your operating system and ensure the NFS service starts at boot.

```
# RHEL 8+
sudo yum install -y nfs-utils

# Ubuntu
sudo apt-get install -y nfs-common
```

NOTE
Reboot your worker nodes after installing the NFS tools to prevent failures when attaching volumes to containers.

iSCSI volumes

Trident can automatically establish an iSCSI session, scan LUNs, discover multipath devices, format them, and mount them to a pod. For ONTAP systems, Trident runs iSCSI self-healing every five minutes to identify, prioritize, and perform repairs that return the current iSCSI session state to the desired state.

Before you begin

- Each node in the Kubernetes cluster must have a unique IQN.
- When using worker nodes that run RHEL/RHCOS with iSCSI PVs, specify the `discard` mountOption in the StorageClass to perform inline space reclamation.
- Ensure you have upgraded to the latest version of `multipath-tools`.

```
# RHEL 8+
sudo yum install -y lsscsi iscsi-initiator-utils device-mapper-multipath
sudo sed -i 's/^\(node.session.scan\).*/\1 = manual/' /etc/iscsi/iscsid.conf
sudo mpathconf --enable --with_multipathd y --find_multipaths n
sudo systemctl enable --now iscsid multipathd
```

```
sudo systemctl enable --now iscsi

# Ubuntu
sudo apt-get install -y open-iscsi lsscsi sg3-utils multipath-tools scsitools
sudo sed -i 's/^\(node.session.scan\).*/\1 = manual/' /etc/iscsi/iscsid.conf
sudo tee /etc/multipath.conf <<-EOF
defaults {
  user_friendly_names yes
  find_multipaths no
}
EOF
sudo systemctl enable --now multipath-tools.service
sudo systemctl enable --now open-iscsi.service
```

Redwash note: The “Before you begin” bullet about `solidfire-san` with Element OS 12.5/12.7 CHAP-algorithm settings was removed (Element off-product). Driver tool-selection lists were trimmed to ONTAP NAS/SAN drivers only.

NVMe/TCP volumes

> **NOTE**
> NVMe requires RHEL 9 or later.

```
# RHEL 9
sudo yum install nvme-cli
sudo yum install linux-modules-extra-$(uname -r)
sudo modprobe nvme-tcp

# Ubuntu
sudo apt install nvme-cli
sudo apt -y install linux-modules-extra-$(uname -r)
sudo modprobe nvme-tcp
```

After installation, verify each node has a unique NQN: `cat /etc/nvme/hostnqn`.

SCSI over FC volumes

You can use the Fibre Channel (FC) protocol with Trident to provision and manage storage resources on an ONTAP system. Configure the WWPNs of the target interfaces and the initiator (host), then configure zoning on the FC switch using the host and target WWPNs. Refer to the ONTAP documentation for *Fibre Channel and FCoE zoning overview* and *Ways to configure FC & FC-NVMe SAN hosts*.

Configure and manage backends

Configure backends

A backend defines the relationship between Trident and a storage system. It tells Trident how to communicate with that storage system and how Trident should provision volumes from it. Learn how to configure the backend for your storage system:

- Configure a backend with ONTAP NAS drivers
- Configure a backend with ONTAP SAN drivers

Redwash note: The following backend chapters were removed entirely as off-product and moved to `TO_DELETE/` : Azure NetApp Files, Google Cloud NetApp Volumes, Cloud Volumes Service for Google Cloud, and NetApp HCI/SolidFire (Element). "Use Trident with Amazon FSx for NetApp ONTAP" was also removed. The only retained backend drivers are `ontap-nas` , `ontap-nas-economy` , `ontap-nas-flexgroup` , `ontap-san` , and `ontap-san-economy` .

ONTAP SAN driver overview

Trident provides the following SAN storage drivers to communicate with the ONTAP cluster. Supported access modes are ReadWriteOnce (RWO), ReadOnlyMany (ROX), ReadWriteMany (RWX), and ReadWriteOncePod (RWOP).

Driver	Protocol	volumeMode	Access modes supported	File systems supported
`ontap-san`	iSCSI, SCSI over FC	Block	RWO, ROX, RWX, RWOP	No filesystem; raw block device
`ontap-san`	iSCSI, SCSI over FC	Filesystem	RWO, RWOP	xfs, ext3, ext4
`ontap-san`	NVMe/TCP	Block	RWO, ROX, RWX, RWOP	No filesystem; raw block device
`ontap-san-economy`	iSCSI	Block	RWO, ROX, RWX, RWOP	No filesystem; raw block device
`ontap-san-economy`	iSCSI	Filesystem	RWO, RWOP	xfs, ext3, ext4

NOTE
Use `ontap-san-economy` only if persistent volume usage count is expected to be higher than supported ONTAP volume limits. Lenovo does not recommend using FlexVol autogrow in ONTAP drivers except `ontap-san`; as a workaround, Trident supports the use of snapshot reserve and scales FlexVol volumes accordingly.

User permissions

Trident expects to run as either an ONTAP or SVM administrator, typically using the `admin` cluster user or a `vsadmin` SVM user, or a user with a different name that has the same role. If you use the `limitAggregateUsage` parameter, cluster admin permissions are required.

Redwash note: The Amazon FSx for NetApp ONTAP user-permission paragraphs (`fsxadmin` account, the FSx `limitAggregateUsage` restriction) were removed (FSx off-product).

Additional considerations for NVMe/TCP

Trident supports the NVMe protocol using the `ontap-san` driver, including IPv6; snapshots and clones of NVMe volumes; resizing an NVMe volume; importing an NVMe volume created outside Trident; NVMe-native multipathing; and graceful or ungraceful shutdown of the Kubernetes nodes. Trident does not support DH-HMAC-CHAP, device mapper (DM) multipathing, or LUKS encryption for NVMe. NVMe is supported only with ONTAP REST APIs (not ONTAPI/ZAPI).

Prepare to configure a backend with ONTAP SAN drivers

For all ONTAP backends, Trident requires at least one aggregate assigned to the SVM. All Kubernetes worker nodes must have the appropriate iSCSI tools installed.

> **NOTE**
> Lenovo DS-Series storage systems differ from DM-Series systems in the implementation of their storage layer. In DS-Series systems, storage availability zones are used instead of aggregates.

Continuation point. The detailed ONTAP SAN and ONTAP NAS backend configuration options, authentication modes, and full YAML examples (all using only `ontap-san*` / `ontap-nas*` drivers) extend here, followed by storage classes, volume provisioning, snapshots, manage/monitor, data protection, security, and the reference chapters. Each follows the same Category B/C rules applied above.

Knowledge and support

Frequently asked questions

This section covers selected redwash decisions for the FAQ. Retained questions keep their upstream answers, rebranded per the Category B rules; the items below were removed or reframed.

Removed FAQ items

Question removed	Reason
Where do I raise a defect?	Routes to NetApp; not Lenovo’s scope
What do I do if I need to raise a request for a new feature?	Routes to NetApp; not Lenovo’s scope
Does Trident support a multi-cloud strategy?	Lenovo storage is on-premises only
Can I use Trident with NetApp Cloud Volumes ONTAP?	Not a Lenovo product
Does Trident support bi-directional CHAP with Element/SolidFire?	No Element on Lenovo

Reframed FAQ items

Now reads as
Does Lenovo support Trident? — Lenovo provides support for Trident when deployed against supported Lenovo DM/DG/DS-Series storage.
Does Trident have any dependencies on other storage products? (vendor-neutral reframe)
How is resource quota translated to a Lenovo storage cluster?

Get help

Lenovo provides support for Trident when deployed against supported Lenovo DM/DG/DS-Series storage.

```
[TODO: Lenovo to provide a description of the support entitlement model and contact
channels — phone, web ticket, email — applicable to Trident on Lenovo storage.]
```

For the support lifecycle, refer to the Lenovo Data Center Support site.

The upstream Trident project maintains a public community on Discord for general questions about the open source project.

Redwash note: Reframed entirely around the Lenovo Data Center Support model. Removed: the `mysupport.netapp.com/…/version-support` link (→ “Refer to the Lenovo Data Center Support site for the support lifecycle”), the `kb.netapp.com/…/Trident_Kubernetes` link, the `cloud.netapp.com/kubernetes-hub` bullet, and the direct `discord.gg/NetApp` link (replaced with prose).

Troubleshooting

Continuation point. The general troubleshooting chapter is largely pass-through (unsuccessful operator/ `tridentctl` deployments, completely remove Trident and CRDs, NVMe node-unstaging and NFSv4.2 notes). Any “send support bundle to NetApp Support” phrasing in AutoSupport context is retained as a Category A exception; non-AutoSupport support routing is changed to Lenovo Data Center Support.

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Redwash note: This is the only file where unmodified NetApp branding is required (Apache 2.0 §6 attribution). The dynamic NetApp legal include (`include::https://raw.githubusercontent.com/NetAppDocs/common/…`) was intentionally removed; the NetApp copyright line is mandatory and retained.

Appendix: Chapters and files removed for the Lenovo deliverable

The following upstream chapters were removed in full (preserved in `TO_DELETE/` for audit, not shipped):

Upstream chapter / files	Reason
Configure backends → Azure NetApp Files (`anf*.adoc`)	Azure NetApp Files — not a Lenovo product
Configure backends → Google Cloud NetApp Volumes (`gcnv*.adoc`)	Google Cloud NetApp Volumes — not a Lenovo product
Configure backends → Cloud Volumes Service for Google Cloud	Cloud Volumes Service — not distributed by Lenovo
Configure backends → NetApp HCI / SolidFire (`element.adoc`)	Element / SolidFire — not a Lenovo product
Use Trident with Amazon FSx for NetApp ONTAP (`trident-fsx*.adoc` , `trident-aws-addon.adoc`)	Amazon FSx / AWS EKS add-on — cloud, off-product
Concept: Volume access groups (`vol-access-groups.adoc`)	SolidFire/Element-only concept
Trident for Docker (`trident-docker/*` , 9 files)	Docker volume plugin (nDVP) — out of Lenovo support scope
Protect applications with Trident Protect (`trident-protect/*`)	Trident Protect — deferred for 25.06
Trident and Trident Protect blogs (`blog-posts.adoc`)	NetApp blog listing
Earlier versions (`earlier-versions.adoc`)	25.06-only release; no past-version index
Controller scalability, Automatic volume expansion, Manage autogrow policies	26.02 features — not in 25.06
RWX NVMe subsystem limits (`trident-nvme-rwx-subsystem-limits.adoc`)	26.02 feature; 404s on the canonical 25.06 site

Redwash legend: highlighted spans mark points where wording was substituted or rewritten from the upstream source; “Redwash note” boxes record eliminations and the rationale, mirroring the per-file `._others.diff` / `._netapp.diff` audit companions.